

Samuel Slone

AI / ML Engineer

(430) 964 0546 — samuel.slone.92@outlook.com — 3813 W San Pedro St, Tampa, FL 33629

Strategic and innovation-oriented **Machine Learning Engineer** and **Data Scientist** with over 9 years of experience in designing, building, and deploying scalable AI systems, with deep domain expertise in **deep learning**, **natural language processing**, **MLOps infrastructure**, and **statistical modeling**; highly skilled in transforming complex **data** into intelligent insights using **Python**, **TensorFlow**, **PyTorch**, and **scikit-learn**, while mastering **deployment** frameworks including **Docker**, **Kubernetes**, **MLflow**, and **Kubeflow** across **cloud** platforms like **AWS** and **Azure**. Adept at managing large-scale datasets using **Spark**, **Snowflake**, and **SQL**, with strong foundations in A/B testing, time-series **forecasting**, and real-time **analytics**. Passionate about aligning **machine learning** systems with business strategy, and building **production**-ready systems that are explainable, reproducible, and performance-optimized.

Skills

- **Languages:** **Python**, **R**, SQL, Java, Bash, JavaScript, Go, Scala, TypeScript
- **ML/DL Frameworks:** TensorFlow, PyTorch, Scikit-learn, Keras, XGBoost, LightGBM, Transformers (Hugging Face), FastText, OpenCV, Dlib
- **MLOps & Lifecycle:** MLflow, Kubeflow, TFX, Airflow, DVC, Docker, Kubernetes, Triton Inference Server, Weights & Biases, TensorBoard
- **Data Engineering:** **Apache Spark**, **Apache Kafka**, Pandas, NumPy, Snowflake, Delta Lake, **Hadoop**, DBT, Great Expectations
- **Cloud Platforms:** AWS (SageMaker, Lambda, EKS, S3, Athena), Azure ML, Azure Databricks, GCP (Vertex AI, **Cloud Functions**, BigQuery)
- **DevOps & Infrastructure:** Git, Jenkins, GitHub Actions, GitLab CI/CD, Terraform, Pulumi, FastAPI, Flask, REST, GraphQL
- **Databases:** PostgreSQL, MySQL, SQL Server, MongoDB, Redshift, DynamoDB, Cassandra, Neo4j
- **Analytics & Visualization:** **Tableau**, **Power BI**, Grafana, Looker, Metabase, Plotly, Dash, Seaborn, Matplotlib
- **Methodologies:** Agile, Scrum, CI/CD, A/B Testing, Experiment Design, Test-Driven Development (TDD), MLOps **Best Practices**

Work Experience

Kiglla - Orlando, FL, US

Machine Learning Scientist / MLOps Engineer (Feb 2023 - Current)

- Led the **architecture** and implementation of an MLOps framework combining **Kubeflow**, **MLflow**, and **Airflow**, allowing seamless **integration** of automated training, validation, drift detection, and CI/CD workflows for both batch and real-time **machine learning** pipelines.
- Engineered and deployed advanced time-series **forecasting** models using **LSTM**, **Prophet**, and **XGBoost**, enabling accurate resource allocation predictions and reducing unplanned downtimes by 23% in industrial **monitoring** use cases.
- Developed streaming AI pipelines using **Apache Kafka** and **Spark Structured Streaming**, providing real-time anomaly detection on high-velocity IoT **data** with millisecond-level latency requirements in **production**.
- Collaborated with cross-functional stakeholders to deliver explainable AI solutions by integrating tools like **SHAP**, **LIME**, and **ELI5** into dashboards, empowering product managers and business analysts to understand and act on model outcomes.
- Designed **production**-grade APIs with **FastAPI**, containerized them using **Docker**, and deployed them to **AWS EKS** with full **monitoring** and autoscaling enabled, ensuring high reliability under dynamic workloads.
- Implemented observability solutions using **Prometheus** and **Grafana**, combined with **Evidently AI** to monitor model metrics such as prediction confidence, drift indicators, and latency, ensuring SLA **compliance** and proactive alerting.
- Built robust batch and streaming **ETL pipelines** using **Apache Spark**, **Snowflake**, and **SQL**, optimizing performance for **data** ingestion across structured, semi-structured, and unstructured sources.
- Conducted end-to-end A/B testing using **Bayesian inference** and hypothesis validation libraries like **SciPy** and **Statsmodels**, ensuring empirical confidence before deploying model upgrades to **production**.

SimpleTalk AI —San Jose, CA , US

Senior **Machine Learning** Engineer(Oct 2021 –Jan 2023)

- Spearheaded the development of **LSTM-based predictive models** focused on estimating SSD endurance and failure rates, which enabled proactive maintenance policies and significantly increased hardware lifespan and reliability across **cloud** deployments.
- Fine-tuned cutting-edge **transformer architectures** such as **BERT** and **GPT**, training on domain-specific corpora to enhance **contextual** understanding for NLP pipelines used in enterprise virtual assistants and support bots.
- Designed highly modular and reusable **machine learning** pipelines using **TensorFlow**, **PyTorch**, and **scikit-learn**, incorporating preprocessing, model training, hyperparameter tuning, and evaluation.
- Deployed and maintained **production** AI services using **Docker**, **Kubernetes**, and **MLflow**, enabling traceable, reproducible, and automatically versioned model deployments with rollback capabilities.
- Integrated model **monitoring** dashboards using **Prometheus** and **Grafana**, allowing real-time tracking of inference performance, latency, precision, and recall in mission-critical NLP services.
- Implemented **reinforcement learning** agents in **OpenAI Gym**, **modeling** dynamic interaction strategies that optimized user-agent communication efficiency in evolving conversation environments.
- Used **Bayesian optimization** and **cross-validation** to iteratively improve model performance, while ensuring robustness across multiple **production** datasets with imbalanced label distributions.
- Leveraged **NVIDIA CUDA** and **TensorRT** to compress and optimize large NLP models for GPU **deployment**, achieving up to 40% reduction in inference latency in **production** workflows.

Sprinklenet -Chicago, IL , US

ML/AI **Data Scientist** (Jan 2018 – Aug 2021)

- Built and maintained complex **machine learning** systems using **TensorFlow**, **Keras**, and **Apache MLlib**, enabling clients to automate fraud detection, recommendation engines, and image recognition workflows across diverse industries.
- Managed the complete model lifecycle using **MLflow** and **DVC**, creating reproducible experiments and audit trails, which streamlined version control and collaboration within distributed teams.
- Utilized **AWS SageMaker** for scalable training, tuning, and endpoint hosting of ML models, integrating with **S3**, **Lambda**, and **CloudWatch** for fully automated **deployment** and **monitoring** pipelines.
- Conducted **big data analysis** using **Snowflake** and **Apache Spark**, extracting insights from datasets exceeding 100 million records and enabling predictive **analytics** dashboards for **customer** churn and marketing performance.
- Migrated legacy **reporting** to modern dashboards using **SQL Server Reporting Services (SSRS)**, improving refresh rates and enabling complex drilldowns and automated visual reports.
- Built AI-backed microservices using **Flask** and **Gunicorn**, containerized with **Docker**, and deployed to **cloud**-native platforms using **AWS ECS** and **CloudFormation** infrastructure templates.
- Developed mobile-first **machine learning** applications using **TensorFlow Lite** and **ONNX**, which enabled offline predictive inference for logistics and **healthcare** companies in field operations.
- Performed A/B testing, feature importance **analysis**, and statistical testing using **SciPy**, **pandas**, and **NumPy**, validating business value of each deployed model before stakeholder release.

Timspark – Fremont, CA, US

Machine Learning Engineer (Jul 2016 – Nov 2017)

- Designed and trained deep **learning** models using **CNNs** and **RNNs** in **TensorFlow** and **Theano** to automate wafer reassembly in semiconductor manufacturing, resulting in over 60% reduction in manual inspection time.
- Built visual anomaly detection systems using **OpenCV**, enabling automatic micro-defect detection from wafer images, significantly improving **production** quality and reducing defect-related returns.
- Created end-to-end **data** pipelines using **Azure Databricks**, aggregating sensor streams into structured formats, with feature engineering and preprocessing routines applied via **PySpark**.
- Applied **Monte Carlo simulations**, **statistical process control**, and outlier detection techniques to model **machine** failure probabilities and implement preventive maintenance alerts.
- Optimized model **deployment** to **NVIDIA GPUs** and **FPGAs**, achieving real-time inference on **production** lines and reducing decision latency to sub-second levels.
- Built interactive dashboards using **Plotly**, **Dash**, and **Flask**, enabling engineering teams to monitor system KPIs and ML insights in real time via mobile and desktop views.
- Contributed to the internal ML framework by creating reusable modules for common model components, improving productivity and consistency across **data science** projects.
- Designed cross-team onboarding guides and Jupyter-based tutorials to help new engineers quickly adopt internal ML **best practices** and infrastructure usage.

Education

Bachelor in Computer Science

University of South Florida| Tampa, FL, US
2012 - 2016

Samuel Slone

- Tampa, FL, US

Contact Information

- nya-yin-ht5@mail.dice.com
- 4309640546

Work History

Total Work Experience: 9 years

- **Machine Learning Scientist** / Mlops Engineer Kiggla
Feb 01, 2023
- **Senior Machine Learning** Engineer Simpletalk AI
Oct 01, 2021
- **ML/AI Data Scientist** Sprinklenet
Jan 01, 2018
- **Machine Learning** Engineer Timspark
Jul 01, 2016

Education

- **Bachelors** | University of South Florida

Skills

- **machine learning** - 10 years
- **matplotlib** - 10 years
- **python** - 10 years

- **apache spark** - 9 years
- **bash** - 9 years
- **docker** - 9 years
- **mentoring** - 9 years
- **mlflow** - 9 years
- **artificial intelligence** - 7 years
- **hypothesis testing** - 7 years
- **power bi** - 7 years
- **scipy** - 7 years
- **snowflake** - 7 years
- **structured query language** - 7 years
- **agile** - 6 years
- **devops** - 6 years
- **fastapi** - 6 years
- **github actions** - 6 years
- **hadoop** - 6 years
- **javascript** - 6 years
- **anomaly detection** - 5 years
- **etl pipeline** - 5 years
- **git** - 5 years
- **grafana** - 5 years
- **java** - 5 years
- **jenkins** - 5 years
- **long short-term memory** - 5 years
- **prometheus** - 5 years
- **scrum** - 5 years
- **typescript** - 5 years
- **amazon athena** - 4 years
- **mongodb** - 4 years
- **postgresql** - 4 years
- **scala** - 4 years
- **terraform** - 4 years
- **amazon elastic kubernetes service** - 3 years
- **apache airflow** - 3 years
- **apache http server** - 3 years
- **apache kafka** - 3 years
- **application programming interface** - 3 years
- **auto scaling** - 3 years
- **autoregressive integrated moving** - 3 years
- **continuous integration** and **continuous** - 3 years
- **data ingestion** - 3 years
- **internet of things** - 3 years
- **kubeflow** - 3 years
- **mlops** - 3 years
- **observability** - 3 years

- **product owner** - 3 years
- **resource allocation** - 3 years
- **statsmodels** - 3 years
- **tableau** - 3 years
- **use cases** - 3 years
- **xgboost** - 3 years
- **data build tool** - 1 years
- **gitlab continuous integration** - 1 years
- **pytorch** - 9 years
- **tensorflow** - 8 years
- **investigation** - 6 years
- **data preprocessing** - 4 years
- **graphics processing unit** - 4 years
- **model deployment** - 4 years
- **bidirectional encoder representations** - 2 years
- **chatbot** - 2 years
- **cloud deployment** - 2 years
- **cognitive services** - 2 years
- **compute unified device architecture** - 2 years
- **fine-tuning** - 2 years
- **generative pre-trained transformer** - 2 years
- **kubernetes** - 2 years
- **language models** - 2 years
- **model development** - 2 years
- **natural language processing** - 2 years
- **openai** - 2 years
- **optimization** - 2 years
- **reinforcement learning** - 2 years
- **scikit-learn** - 2 years
- **time tracking** - 2 years
- **flask** - 6 years
- **numpy** - 6 years
- **amazon cloudwatch** - 4 years
- **amazon elastic container service** - 4 years
- **amazon sagemaker** - 4 years
- **amazon simple storage service** - 4 years
- **amazon web services** - 4 years
- **amazon web services lambda** - 4 years
- **apache spark mllib** - 4 years
- **aws cloudformation** - 4 years
- **cloud native** - 4 years
- **collaboration** - 4 years
- **computer vision** - 4 years
- **continuous deployment** - 4 years
- **data analysis** - 4 years

- **fraud detection** - 4 years
- **keras** - 4 years
- **microservices** - 4 years
- **pandas** - 4 years
- **predictive analytics** - 4 years
- **report generation** - 4 years
- **sql server reporting services** - 4 years
- **convolutional neural networks** - 2 years
- **data science** - 2 years
- **databricks** - 2 years
- **deep learning** - 2 years
- **fabrication** - 2 years
- **feature engineering** - 2 years
- **field-programmable gate array** - 2 years
- **good practice** - 2 years
- **interactive dashboards** - 2 years
- **jupyter notebook** - 2 years
- **onboarding** - 2 years
- **opencv** - 2 years
- **performance metrics** - 2 years
- **plotly** - 2 years
- **process control** - 2 years
- **recurrent neural network** - 2 years
- **simulation** - 2 years

Work Preferences

- Desired Work Settings: Remote
- Likely to Switch: True
- Willing to Relocate: False
- Travel Preference: 0%
- Work Authorization:
 - US
- Work Documents:
 - US Citizen
- **Security** Clearance: True
- Desired Hourly Rate: 60+ (USD)
- Desired Salary: 100,000+ (USD)
- Third Party: False
- Employment Type:
 - Full-time
 - Part-time

- Contract - Corp-to-Corp
- Contract - Independent
- Contract - W2
- Contract to Hire - Corp-to-Corp
- Contract to Hire - Independent
- Contract to Hire - W2

Profile Sources

- linkedin: <https://linkedin.com/in/samuel-slone-s1991s0301>
- Dice: <https://www.dice.com/employer/talent/profile/2e7cd592e74ffef9f0b948942db797ed>